Air Quality Index Prediction and Attention Faithfulness Across Eight Indian Cities

Full project report | Methods, results and discussion

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Abstract

This project investigates station-level air quality index (AQI) prediction in eight Indian cities and asks whether attention weights in multimodal models faithfully reflect model behavior. It combines CPCB-reported station AQI with satellite, reanalysis and archived forecast-weather data. Two practical tasks are separated: retrospective same-day estimation and one-day-ahead forecasting. Models are evaluated against persistence on chronological held-out years; a separate attention suite is tested for accuracy, cross-city transfer and explanation faithfulness. On paired historical cohorts, a history-aware Ridge model reached 32.806 RMSE in 2024 same-day estimation, a 28-day GRU reached 32.516 RMSE in 2025 next-day forecasting, and a weather-assisted XGBoost candidate reached 30.270 RMSE on January-September 2026. The 2026 candidate lost sharp-change precision against the GRU and was therefore not promoted. Attention improvements were architecture-dependent, and attention rankings agreed less with fixed-model input-replacement sensitivity than SHAP in the sampled audits. The reserved October-December 2026 prospective study has no valid score as of this report date.

Keywords: air quality index, time-series forecasting, spatiotemporal learning, GRU, XGBoost, attention faithfulness, cross-city transfer

1 Introduction

AQI varies across stations, seasons and weather regimes. A graph of tomorrow's estimate can appear convincing simply because yesterday's AQI is strongly related to today's. The project therefore compares every operational candidate with an exact previous-calendar-day persistence reference. Its research gap goes beyond error reduction: attention maps are often presented as explanations without testing whether the weighted inputs actually affect a trained model's output. The study tests predictive performance and explanation faithfulness as distinct questions [1], [2].

The implemented system has a practical dashboard and a research evaluation. The dashboard demonstrates historical station and city predictions, selectable modes, paired observed-versus-predicted plots and model comparisons. The research suite evaluates uniform, temporal, feature and spatial attention controls. A public static demonstration displays saved historical results; a local package supports custom inputs. Neither static historical curve is a newly issued live forecast.

2 Objectives and scope

Objective	How it is tested
Estimate and forecast station AQI	Separate same-day estimation from next-day prediction; compare with persistence on matched rows.
Assess generalization	Hold out full future-year periods and evaluate leave-one-city-out transfer in a separate research suite.
Audit explanations	Compare attention rankings with SHAP and fixed-model replacement sensitivity; inspect deletion behavior.
Make outcomes demonstrable	Provide city and station controls, interactive date-filtered graphs, hover values and downloadable saved results.

3 Data and preparation

The configuration covers 107 CPCB monitoring stations in Bengaluru, Chennai, Delhi, Hyderabad, Jaipur, Kolkata, Lucknow and Mumbai. CPCB hourly reported AQI supplies the response and historical lags. Sentinel-5P NO2 and HCHO columns, MODIS MAIAC AOD, ERA5-Land reanalysis weather and station coordinates support the retrospective and attention studies. The separate 2026 forecast candidate uses earlier-issued, city-scale GFS Seamless weather from Open-Meteo's Previous Runs API [3]-[6]. Satellite columns and AOD are remote-sensing indicators, not direct station measurements of ground-level pollutant concentration.

Raw CPCB timestamps are converted from IST to UTC. The daily response is the quality-filtered reported AQI endpoint at 23:30 UTC on target date d, corresponding to 05:00 IST the following morning. A station-day needs at least 12 valid hourly readings, an endpoint in the 0-500 range and no flagged stuck series. Missing targets are left missing. The end-of-day endpoint avoids blurring adjacent days by averaging overlapping hourly trailing-24-hour AQI values. Following-month files can be needed to establish month-end endpoints [7].

Environmental records are sampled on city grids and joined to each station's nearest grid point by date. Features include transformed temperature, humidity, wind, pressure and rainfall, satellite columns, location, seasonal terms and exact calendar AQI lags. An absent calendar day remains absent rather than becoming a false 'yesterday' value. Satellite missingness is retained; training-only imputation, indicators and scaling are used as appropriate. Target-day observed environment is allowed only for the retrospective same-day task. The next-day models exclude target-day observed AQI, satellite retrievals and realized weather [7].

4 System and model design

4.1 Same-day estimation

A chronological 2023 selection chose a Ridge regression model with AQI history for retrospective same-day estimation. Comparators include persistence, environment-only Ridge, history-aware tree ensembles and an environment-only GRU. The task can use target-day environmental observations and is not labeled an operational forecast. The environment-only comparator tests how much historical AQI contributes to accuracy [7].

4.2 One-day-ahead GRU

The default next-day model encodes the preceding 28 calendar days of AQI, availability, date position and station coordinates. One GRU layer with 64 hidden units feeds a dropout and dense regression head. Three seeds are averaged. Training uses CUDA, internal chronological early stopping, a 27-day boundary purge and preprocessing fitted only on the permitted training block. A prediction issued for date d uses history ending at d - 1; the target value is absent [7], [8].

4.3 Weather-aware candidate

An XGBoost residual model predicts a correction to previous-day AQI. It combines 28 AQI lags and rolling statistics with station location, seasonality and five daily archived forecast-weather variables. Complete 24-hour earlier-lead forecasts are required; missing forecasts are not replaced with realized target-day weather. Open-Meteo's previous_day1 fields correspond to values predicted 24 hours before their valid times, but the archive does not certify original public ingestion latency for every historical value [6], [7], [9].

4.4 Attention and explanation experiments

The research models compare learned temporal attention (TA), feature attention (FA), spatial attention (SA) and combinations with within-family uniform-weight controls. A saved checkpoint is then held fixed while inputs are replaced according to ranked groups. Spearman agreement, deletion effects and SHAP comparisons measure whether rankings align with model output sensitivity. This audit describes model behavior under the chosen intervention; it is not a causal study of atmospheric pollution sources [1], [2], [10].

5 Evaluation protocol

Historical task	Fit	Select / calibrate	Held-out test
2024 same-day	Jan-Sep 2023	Oct / Nov-Dec 2023	2024; 28,803 station-days
2025 next-day	Jan 2023-Sep 2024	Oct / Nov-Dec 2024	2025; 30,263 station-days
2026 weather comparison	Jan 2024-Sep 2025	Oct / Nov-Dec 2025	Jan-Sep 2026; 20,086 paired station-days

Table 1. The three test cohorts have different periods and eligible rows; comparisons are valid within a row, not across years.

Root mean squared error (RMSE) is the principal model-selection measure; mean absolute error (MAE), R-squared, bias and interval coverage supplement it. A positive paired RMSE gain is reference RMSE minus candidate RMSE. Uncertainty for the paired 2026 gains uses 2,000 calendar-week block bootstrap draws, retaining all station observations in each sampled week. Nominal 90% station prediction intervals are set from a separate calibration block. Future-year test labels are not used for model fitting, selection or calibration [7].

The 2026 comparison uses exactly the same 20,086 station-days for weather XGBoost, the frozen GRU and persistence, spanning eight cities and 91 of 107 configured stations. The verifier recomputed metrics and city groups, reloaded model outputs, checked hashes, interval bounds, exact-day references and station exclusions. The transfer experiment instead holds out one city at a time and is not a prospective future-year forecast test [7].

6 Results

6.1 Historical prediction accuracy

Task and model	Rows	MAE	RMSE	R^2
2024 same-day: history Ridge	28803	23.001	32.806	0.892
2024 same-day: persistence	28803	23.716	36.197	0.869
2024 same-day: env-only Ridge	28803	45.804	63.239	0.599
2025 next-day: 28-day GRU	30263	21.857	32.516	0.887
2025 next-day: persistence	30263	22.981	35.388	0.867
2026 next-day: weather XGBoost	20086	19.345	30.270	0.848
2026 next-day: frozen GRU	20086	20.394	31.707	0.833
2026 next-day: persistence	20086	21.763	34.668	0.800

Table 2. Saved held-out historical metrics; errors are AQI points. Within each year, models share eligible rows.

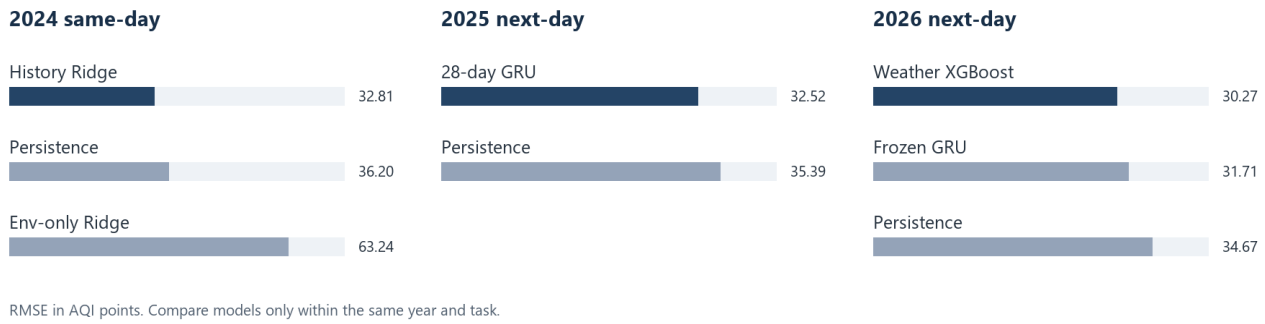


Fig. 1. RMSE of matched models within each distinct historical evaluation cohort.

In 2024, history-aware Ridge reduced RMSE by 3.391 AQI points (9.4%) relative to persistence; its paired 95% block-bootstrap interval was 2.025-4.671. In 2025, the GRU reduced RMSE by 2.872 points (8.1%), with a paired 95% interval of 2.168-3.605. Its nominal 90% station interval covered 95.36% of test observations. These are historical results, not guaranteed live performance [11], [12].

On paired January-September 2026 rows, weather XGBoost lowered RMSE by 1.437 points relative to the frozen GRU (95% weekly-bootstrap interval 0.376-2.614) and by 4.398 points relative to persistence (3.190-5.715). It had lower RMSE in each represented city. Its nominal 90% interval covered 94.88% of the historical test rows. The test included 91 stations; missing exports and daily quality failures limit coverage [13].

6.2 Abrupt changes and high AQI

2026 measure at 50 AQI threshold	Weather XGBoost	Frozen GRU
Observed sharp changes	2315	2315
Forecast-flagged changes	799	268
Correct directional detections	422	154
Recall	18.23%	6.65%
Precision	52.82%	57.46%

Table 3. A detection requires a forecast change of at least 50 in the observed direction.

Weather XGBoost detected more abrupt changes but issued more alerts. Its recall rose from 6.65% to 18.23%, while precision fell from 57.46% to 52.82%. The predeclared promotion rule required positive paired RMSE improvement against GRU and persistence without losing either sharp-change recall or precision. Because precision fell, the GRU remains the default. The weather model's mean error was -38.51 AQI points for observed AQI 301-400 and -44.36 for 401-500 (94 observations in the latter band), so lower overall RMSE did not remove severe-day weakness [13], [14].

6.3 Attention, transfer and explanation audit

Experiment	Control	Attention variant	Result
2023 sequence chrono	Uniform RMSE 78.361	TA RMSE 84.394	No accuracy gain
2023 spatial chrono	Uniform RMSE 75.328	SA RMSE 65.836	Architecture-specific gain
8-city LOCO sequence	City-macro RMSE 55.11	TA 55.23	No city-macro gain
8-city LOCO spatial	City-macro RMSE 57.19	TA+SA 54.16	Gain in 5 of 8 cities

Table 4. Attention is compared only with the uniform control in the same architecture and protocol.

The corrected 2023 chronological attention comparison used 5,434 station-days. The corrected LOCO transfer study used eight held-out cities and 29,923 station-days. Equal-city weighting matters: sequence TA had a marginally lower pooled RMSE (71.61 versus 71.67 for Uniform) but a slightly worse city-macro RMSE (55.23 versus 55.11). The spatial combination improved city-macro RMSE from 57.19 to 54.16, yet three cities did not improve [14], [15].

In fixed-model audits of 300 sampled rows per corrected FA and SA checkpoint, row-weighted mean Spearman agreement between attention ranking and input-replacement sensitivity was below SHAP's in all four comparisons. Under mean replacement, FA attention/SHAP correlations were 0.420/0.664; spatial SA correlations were 0.242/0.419. Under same-city seasonal donors, the corresponding pairs were 0.392/0.546 and 0.483/0.586. The change with replacement method shows that this is a limited intervention-based result, not proof that SHAP or attention reveals physical causation. The LOCO checkpoints have not received the same faithfulness audit [14].

6.4 Prospective status

The reserved prospective period is 13 October-31 December 2026. At the latest saved check (2026-10-24T07:26:37.075221+00:00), the October CPCB request had 0 of 107 station exports, 0 usable new station-days and 0 immutable before-target forecast days. The weather snapshot for 2026-10-25 had 0 of 8 complete cities. Consequently there is no valid prospective RMSE, MAE, promotion decision or final evaluation. The final scoring gate opens no earlier than 1 January 2027 and also needs quality-checked January boundary readings [16].

7 Discussion

7.1 Why some models performed better

Persistence was a difficult benchmark because the previous day's AQI contains much of the current level. The 2024 environment-only Ridge error was nearly twice the history-aware Ridge error, supporting the value of an AQI anchor in retrospective estimation. The 2025 GRU could use a longer exact-day sequence and improved average error over a one-day repeat; this comparison does not isolate which channel produced the gain. The 2026 residual XGBoost explicitly starts from yesterday's level and learns nonlinear corrections using historical and forecast-weather context. Its paired gain is a model-level observation; because architecture and inputs differ, it cannot be interpreted as the causal effect of weather alone [11]-[13].

7.2 Why average error was insufficient

Abrupt pollution changes are uncommon relative to ordinary days and are harder to predict. The weather candidate moved farther from persistence more often than the GRU, increasing true detections but also false alerts. Its high-AQI underprediction matters even when pooled RMSE is lower. Pooled station-day metrics also weight heavily observed cities more strongly; incomplete exports may alter which stations are represented. These are reasons to preserve the predeclared safety-oriented promotion rule and present city and event metrics beside aggregate RMSE [13], [14].

7.3 What attention results mean

Learned attention helped some spatial models but not the tested sequence model. That variation is consistent with attention serving as a flexible learned aggregation method rather than a universal accuracy improvement. The ranking audit directly challenges the assumption that a large attention weight identifies an input that most changes output. Replacement interventions can be unrealistic when weather variables are correlated, and their results changed with donor choice. The current evidence supports cautious model-behavior descriptions, not a claim about the real-world causes of AQI [1], [14], [15].

8 Dashboard and demonstration

The demo allows a viewer to switch between same-day estimates and historical next-day backtests, choose a city or station, inspect predicted and observed AQI by date, show or hide graph series and read exact hover values. Its comparison table summarizes model error and coverage, while station rankings and downloadable filtered data permit a closer look. City curves are equal-weight means of available paired stations, not official published city AQI. The static public page shows saved historical evidence; custom supplied AQI histories are evaluated through the local dashboard package. These user-entered values are demonstration inputs, not independently verified live observations [17].

9 Limitations and future work

The main limitations are uneven station coverage, absent endpoint readings, cloud-related satellite gaps, city-scale rather than station-specific archived weather, uncertain historical source publication latency and weak detection of sudden or very high AQI. The faithfulness audits sample limited rows and use replacement interventions; LOCO model accuracy does not validate LOCO explanation faithfulness. The prospective record is currently empty. The correct next study is to capture eligible prior-day AQI and complete earlier-issued weather before each target boundary, preserve immutable forecasts, then quality-check subsequent CPCB observations and evaluate the frozen models once. Further improvements should focus on event recall/precision, station availability, calibration under shift and explanation stability across cities, using fresh development data before another independent test [14], [16].

10 Conclusion

The project delivers an eight-city AQI research pipeline and an interactive demonstration grounded in saved, verified historical experiments. History-aware Ridge improved 2024 same-day estimation over persistence; the 28-day GRU improved 2025 next-day error; weather-assisted XGBoost improved average 2026 historical error but did not meet the abrupt-change precision condition for promotion. Spatial attention showed selective cross-city accuracy gains, while sampled faithfulness checks did not support treating attention weights as automatically reliable explanations. The frozen GRU remains the next-day default. No claim of validated October-December 2026 prospective performance is possible until before-target records and ground observations exist.

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